LAB 4 — CONNECTIVITY OVERVIEW

Front and rear are the terminated ends. Cockpit taps both backbones.
Each zone has its own two isolation groups; termination remains on the bus side.
Input fuse is not per-motor current limiting. See Fuse-and-scope-review.md.

Zone sheets

Front zone



File: Front.kicad_sch

Cockpit zone



File: Cockpit.kicad_sch

Rear zone



File: Rear.kicad_sch

Power sheet

Power



File: Power.kicad_sch

Design notes

CAN_A_H/L and CAN_B_H/L are separate root nets. Same-name local labels on this root join the shown sheet pins. Child labels connect only through those ports.

Power flags document external battery power and outputs after passive filters; they do not prove current capacity, regulation, thermal behavior or PCB completeness.

See November2-Anson-review.md for review changes and verification.

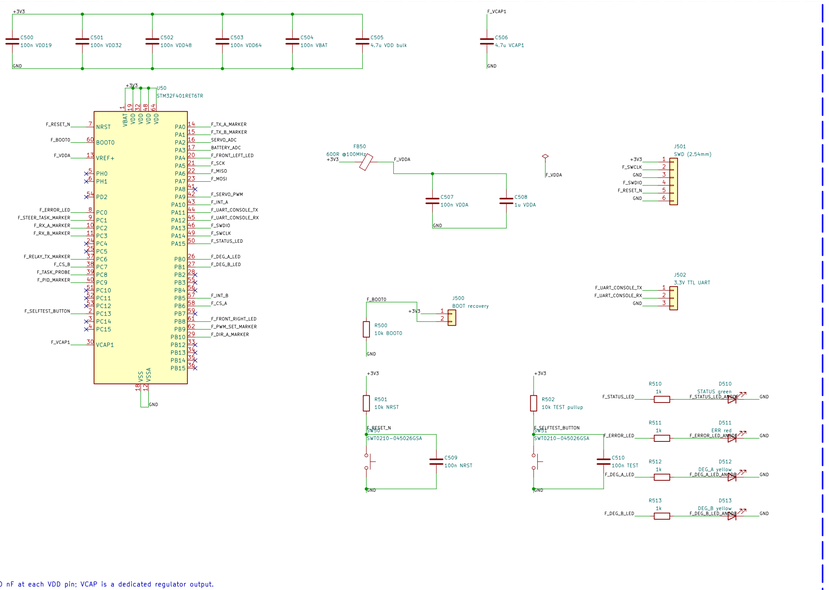
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Sheet: /
File: Lab4.kicad_sch

Title: Lab 4 — three-zone controller

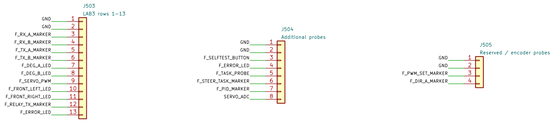
Size: A2	Date: 2026-11-02	**Rev: Review 2F**
KiCad E.D.A. 10.0.4		Id: 1/5

Microcontroller — STM32F401



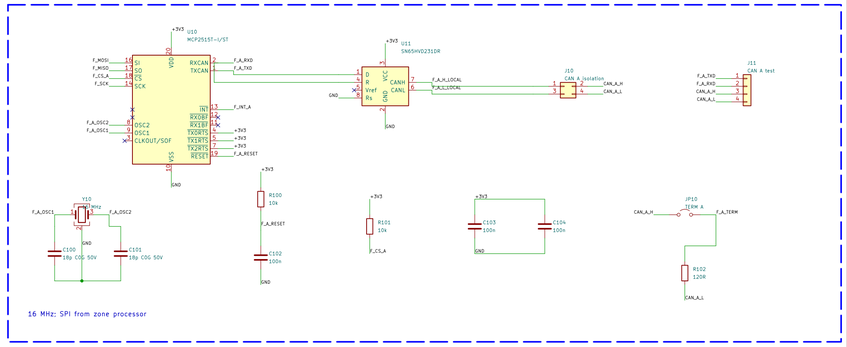
100 nF at each VDD pin; VCAP is a dedicated regulator output.

Accessible test headers



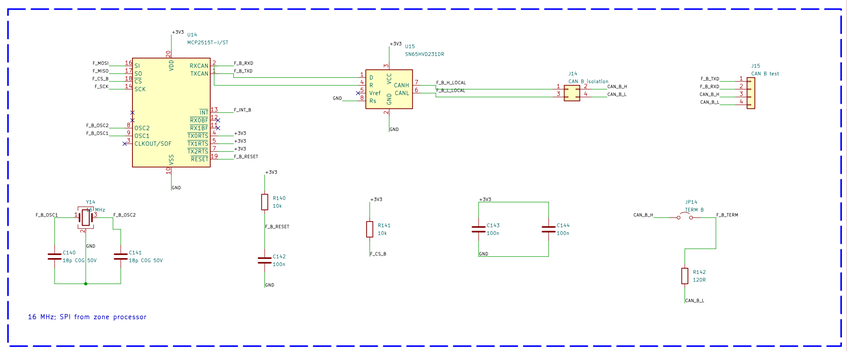
Lab 3 pin assignments retained. Do not apply 5 V to MCU/ADC inputs.

CAN A



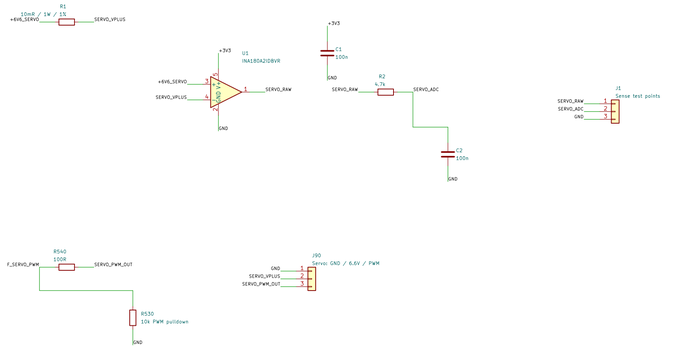
16 MHz SPI from zone processor

CAN B

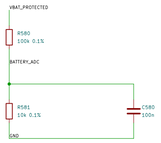


16 MHz SPI from zone processor

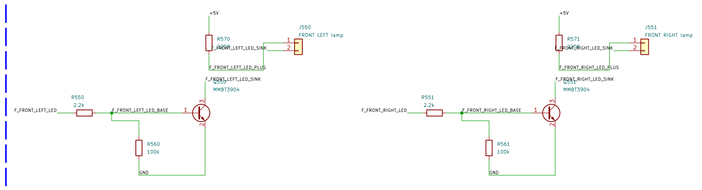
Servo supply, high-side current sense and PWM



Battery voltage monitor



Vehicle lamps



Sheet connections

+3V3 · +5V · +6V6_SERVO · SERVO_VPLUS · GND · CAN_A_H · CAN_A_L · CAN_B_H · CAN_B_L · VBAT_PROTECTED

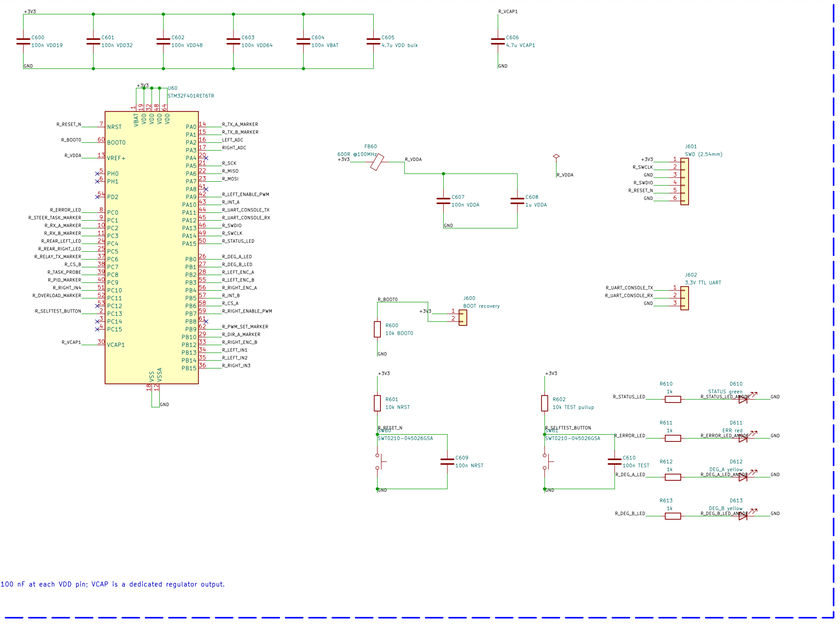
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Sheet: /Front zone/
File: Front.kicad_sch

Title: Front: steering, telemetry and dual CAN

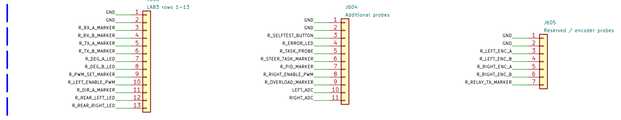
Size: A1	Date: 2026-11-02	Rev: Review 2F
KiCad E.D.A. 10.0.4		Id: 2/5

Microcontroller — STM32F401

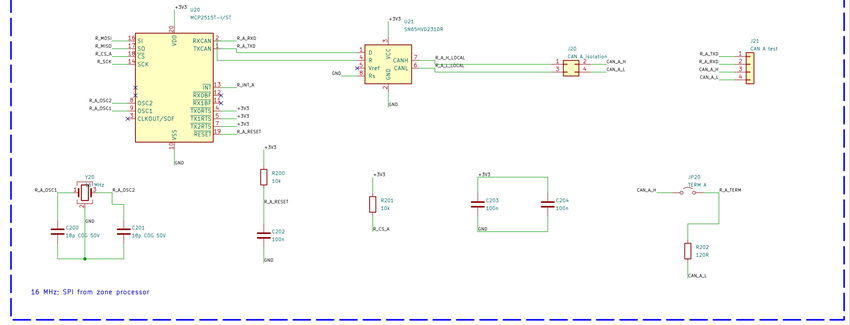


100 nF at each VDD pin; VCAP is a dedicated regulator output.

Accessible test headers

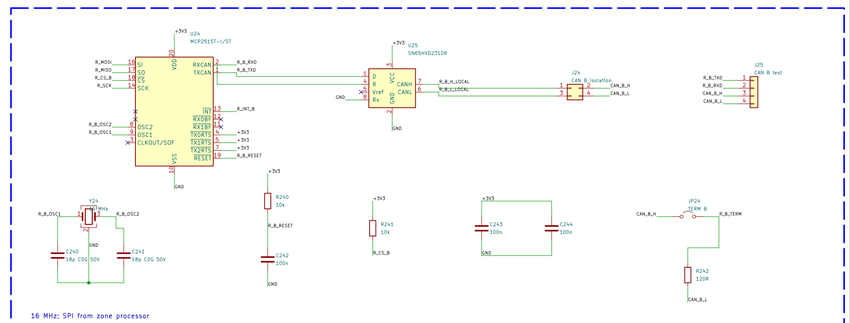


CAN A



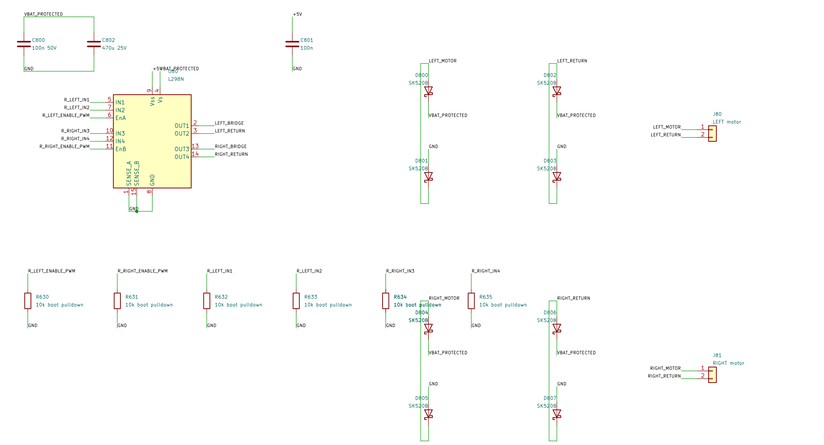
16 MHz; SPI from zone processor

CAN B



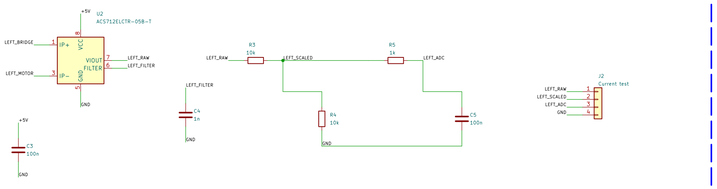
16 MHz; SPI from zone processor

Motor H-bridge / output clamps / connectors

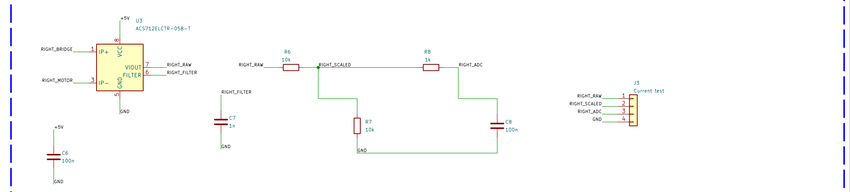


All flyback clamps are on the motor side of the ACS712.
Recirculation can bypass the sensor; telemetry is not complete phase current.

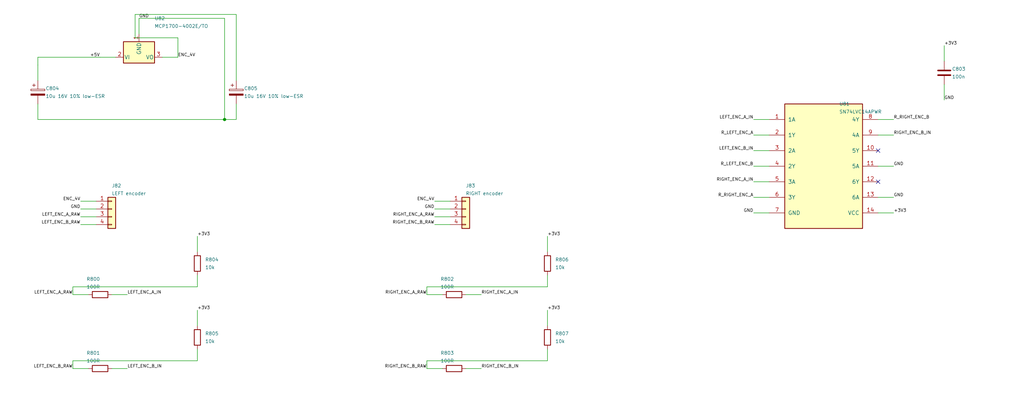
Left motor current / ADC filter



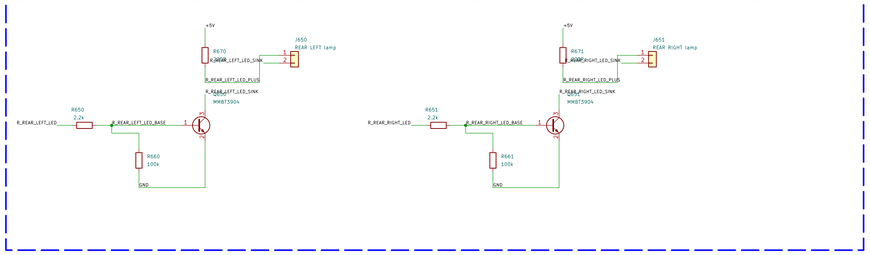
Right motor current / ADC filter



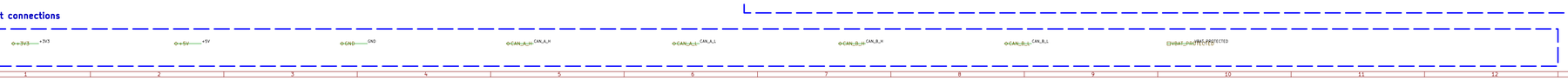
Encoder power / 5 V-tolerant Schmitt inputs



Vehicle lamps



Sheet connections



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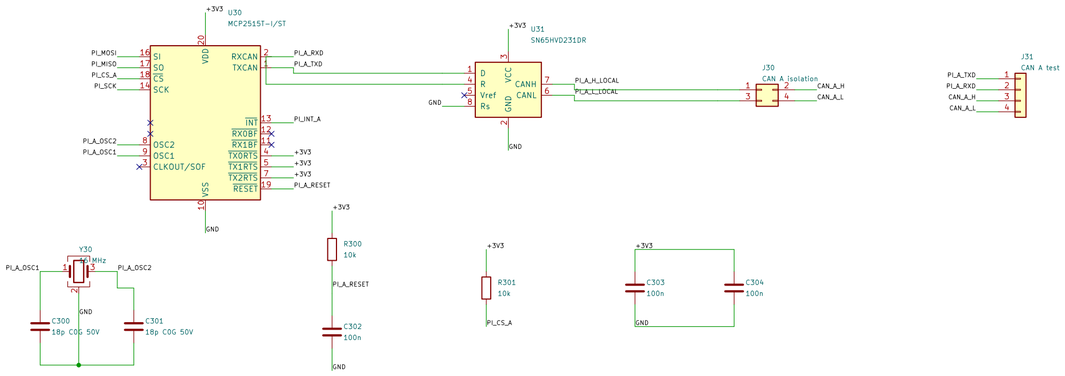
Sheet: /Rear zone/

File: Rear.kicad_sch

Title: Rear: drivetrain, encoders and dual CAN

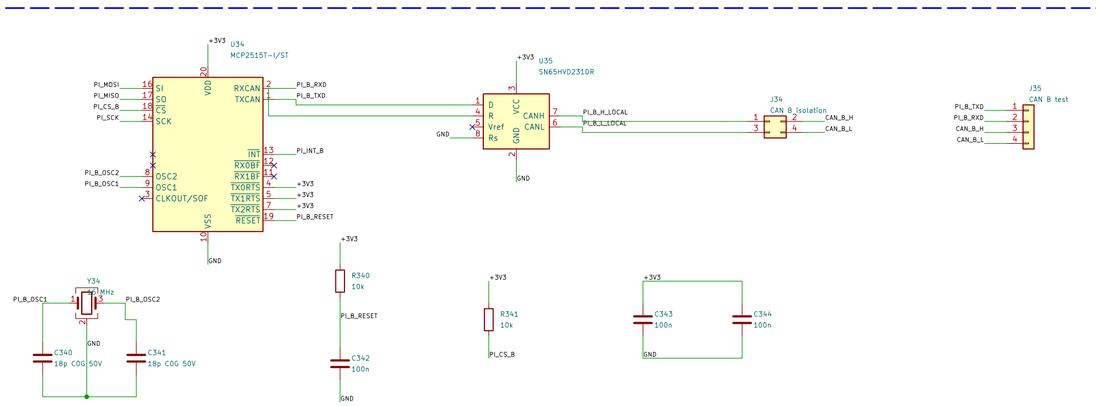
Size: A1	Date: 2026-11-02	Rev: Review 2F
KiCad E.D.A. 10.0.4		Id: 3/5

CAN A



16 MHz; SPI from zone processor

CAN B

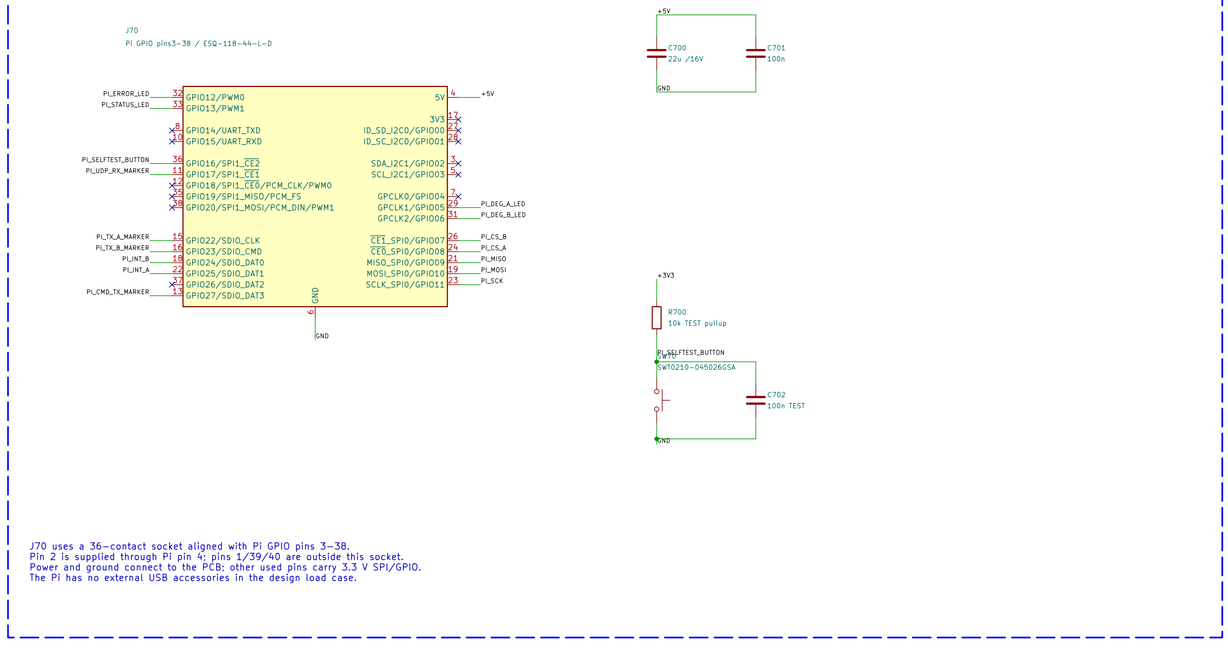


16 MHz; SPI from zone processor

CAN Isolation

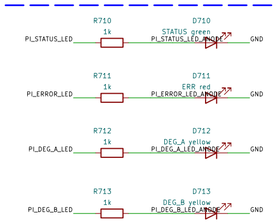
CAN termination is at the front and rear zones only.
Remove both isolation shunts to disconnect one local transceiver.

Pi interface: physical header pins 3–38



J70 uses a 36-contact socket aligned with Pi GPIO pins 3–38.
Pin 2 is supplied through PI pin 4; pins 1/39/40 are outside this socket.
Power and ground connect to the PCB; other used pins carry 3.3 V SPI/GPIO.
The Pi has no external USB accessories in the design load case.

Status LEDs



Accessible test headers



Sheet connections

+3V3 +3V3 · +5V +5V · GND GND · CAN_A_H CAN_A_H · CAN_A_L CAN_A_L · CAN_B_H CAN_B_H · CAN_B_L CAN_B_L

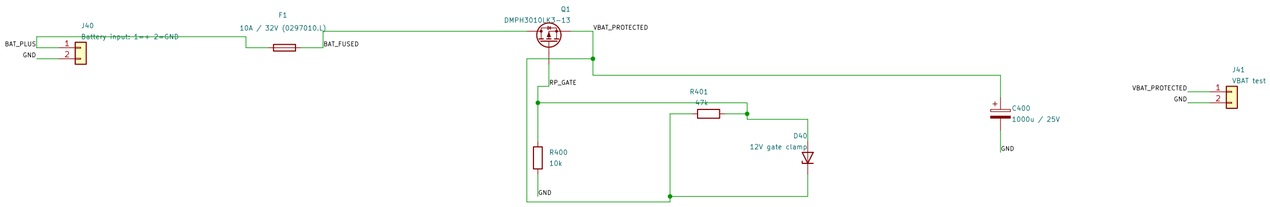
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Sheet: /Cockpit zone/
File: Cockpit.kicad_sch

Title: Cockpit: Raspberry PI interface and dual CAN

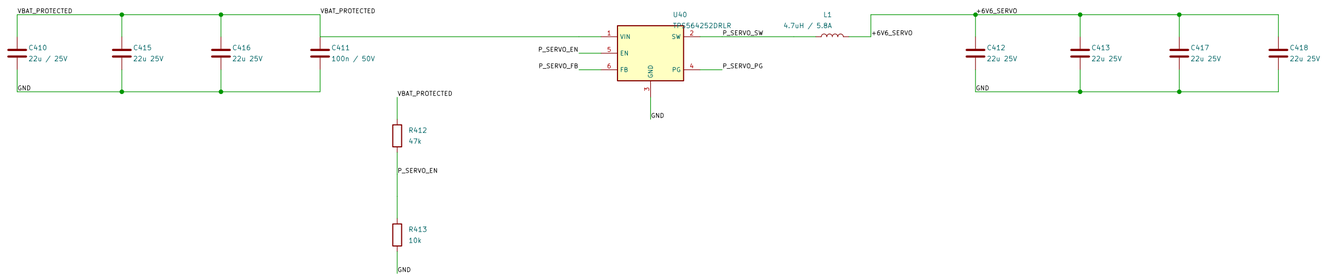
Size: A2	Date: 2026-11-02	Rev: Review 2F
KiCad E.D.A. 10.0.4		Id: 4/5

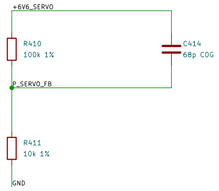
3S LiPo input / fuse / reverse-polarity protection



F1: selected 10 A MINI fuse; holder 3568. Input-path fuse, not per-motor current limiting.
Verify the installed fuse and harness before battery operation; see Fuse-and-scope-review.md.

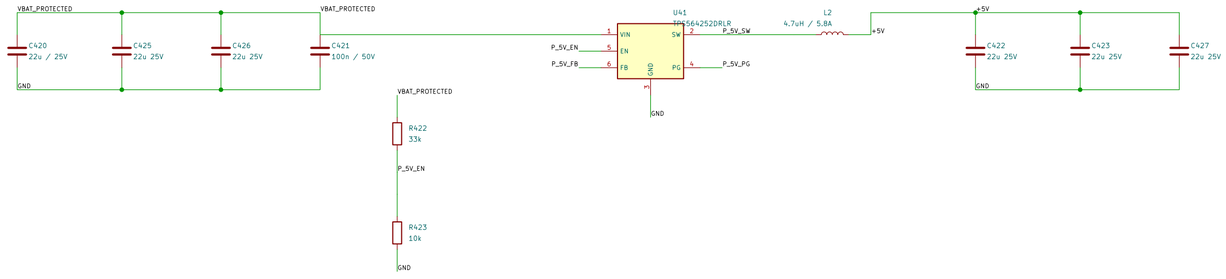
Servo rail: 6.6 V

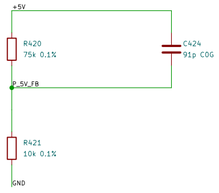






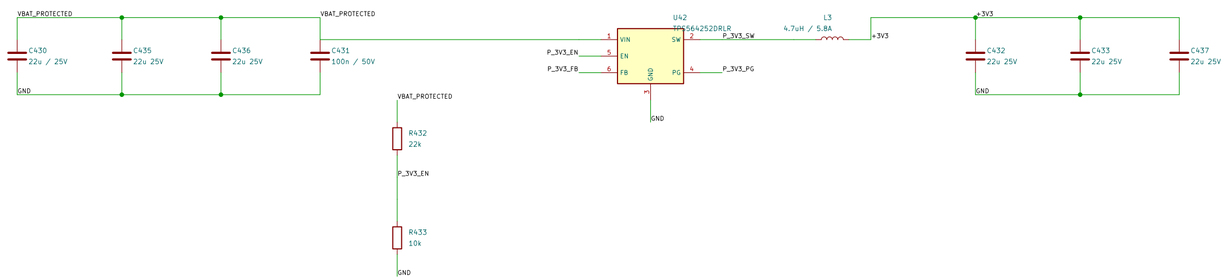
Pi / L298 logic rail: 5.1 V

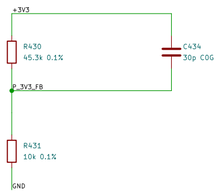






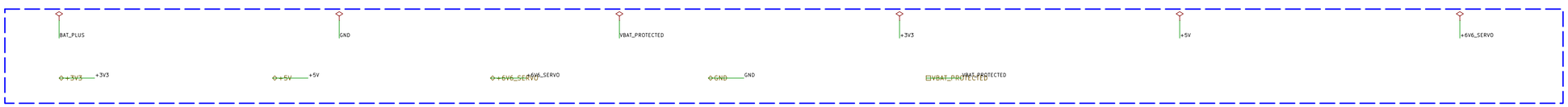
Logic rail: 3.3 V







Power flags / sheet connections



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Sheet: /Power/
File: Power.kicad_sch

Title: Power: input protection and regulated rails

Size: A2	Date: 2026-11-02	**Rev: Review 2F**
KiCad E.D.A. 10.0.4		Id: 5/5